

Trust Decay Agent: Autonomous IoT Security Through Continuous Trust Modelling



Why Today's IoT Security Breaks Down

Gradual Compromises Unnoticed

Subtle, slow-developing threats often bypass traditional, static security thresholds.

Binary Alerts Too Late

Alerts are typically triggered only after a significant breach, offering reactive instead of proactive defence.

Human Responses Don't Scale

Reliance on manual intervention for every alert is unsustainable given the sheer volume of IoT devices and data.

Rethinking IoT Security: The Dynamic Nature of Trust

Security in the IoT landscape demands a more adaptive approach than mere detection. Trust is not a static state; it is a dynamic attribute that evolves with device behaviour.

- Security is more than detection
- Trust evolves with behaviour
- Trust decays and recovers over time



The Trust Decay Agent: Our Autonomous Solution



Continuous Evaluation

Continuous behavioural analysis ensures immediate threat detection.



Autonomous Decisions

AI-driven logic enables devices to make security decisions without human oversight.



Edge-Native Design

Optimised to operate directly on IoT devices, minimising latency and cloud reliance.

How It Works: Simple, Autonomous, Edge-First

1

Observe

Lightweight behavioural signals are continuously monitored from IoT devices.

2

Reason

Continuous trust decay logic evaluates these signals to determine trust levels.

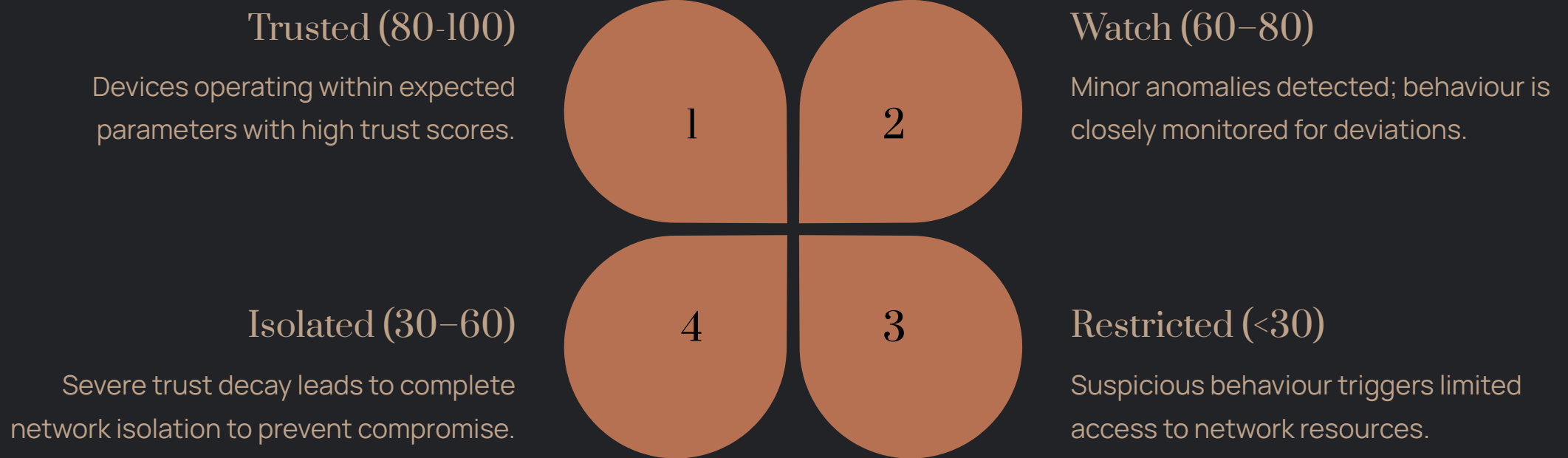
3

Act

Graduated security responses are implemented based on the evolving trust model.

Observe → Reason → Act runs continuously at the edge.

Trust in Motion: Decay and Recovery



Autonomous Action: From Trust to Response



- **Monitoring** Continuous oversight of device activities and network interactions.
- **Traffic Shaping** Adjusting data flow to control potential threats and maintain network stability.
- **Network Isolation** Segregating compromised devices to prevent lateral movement of threats.
- **Quarantine** Complete containment of malicious devices for further analysis and remediation.

Responses escalate automatically as trust decays.

Why Trust Decay Beats Traditional Security

Trust Decay Agent

- Continuous trust assessment
- Autonomous response mechanisms
- Edge-native processing for low latency



Traditional Security

- Binary, static alerts
- Human-dependent interventions
- Cloud-reliant infrastructure



A Better Foundation for IoT Security

Scales Naturally

Designed to effortlessly manage an expanding number of IoT devices.

Thrives on Uncertainty

Adapts to evolving threat landscapes and unpredictable device behaviours.

Fits IoT Constraints

Operates efficiently within the resource limitations of IoT environments.

Seamless Integration

Easily integrates with existing IoT infrastructures without extensive overhaul.

From Detection to Autonomous Decision-Making

- By Chitesh Marmat

